CSI SAFE

Software Course | Duration: 5 Weeks

A focused course on slab and foundation design using CSI SAFE, covering RC and post-tensioned slab systems, punching shear, and mat/raft foundation design commonly required in high-rise and industrial projects.

Who Should Enroll: Structural design engineers and STAAD/ETABS users who want to specialize in slab and foundation design.

Prerequisites

- Basic RCC design knowledge
- Familiarity with STAAD.Pro or ETABS is an advantage

Learning Outcomes

- Model slab, beam, and foundation systems with correct grid and property setup
- Apply realistic loading for flat slabs, PT slabs, and rafts
- Run and interpret meshed finite-element slab analysis
- Perform punching shear checks and IS 456-based slab design
- Design a mat/raft foundation with reinforcement detailing

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	**Modeling Fundamentals**	– Interface overview, grid and structural layout setup – Slab, beam, and column property definitions
Week 2	**Loading & Slab Types**	– Load assignment: dead, live, finishes, partition loads – Modeling conventional RC slabs vs post-tensioned slabs
Week 3	**Analysis**	– Mesh generation strategy and analysis run – Reading moment/shear contours and deflection results
Week 4	**Design Checks**	– Punching shear checks around columns – Slab and mat foundation design per IS 456 – Deflection and crack-width verification
Week 5	**Capstone Project**	– Design of a flat-slab or raft foundation system for a sample building – Design report with reinforcement detailing output

Capstone Project

Design of a flat slab / raft foundation for a sample multi-storey building, including a full design report.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, along with a portfolio-ready project file that can be showcased to employers or clients.